



PROFESSIONAL CAREER PROGRAM

Multi-Cloud Data Engineer Program

(GCP + AZURE)

with Case Studies and Real-world Projects

STRUCTURED LEARNING • HANDS-ON LABS • INDUSTRY PROJECTS



COURSE TRAINER
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Course Snapshot

EVERYTHING AT A GLANCE

ROLE

Multi-Cloud Data Engineer

DURATION

4 months

MODE

Online

LANGUAGE

English

COURSE ROADMAP

Your Learning Journey

A structured path from cloud foundations to production-ready data engineering across Google Cloud and Microsoft Azure — reinforced with case studies and real-world projects.

01 Cloud Introduction

Cloud computing fundamentals and why organizations move to the cloud — GCP, Azure & AWS.

02 Foundations

GCP Cloud Basics · Linux Basics on Cloud Shell · Python for Data Engineer.

03 GCP Data Engineering Tools

Cloud Storage · Cloud SQL · BigQuery · DataProc · DataFlow · Pub/Sub · Cloud Composer.

04 Azure Data Engineering Tools

Azure Introduction · ADLS · Azure SQL DB · Azure Data Factory · Azure Synapse.

05 Databricks on Azure / GCP

Lakehouse, Unity Catalog, Delta Lake, ELT pipelines, streaming & Autoloader case studies.

06 Outcomes & Interview Readiness

End-to-end pipelines, Medallion architecture, and scenario-based interview preparation.

01 Foundations

Cloud, GCP platform, Linux & Python essentials for data engineers

01 Cloud Introduction

4 TOPICS

- What is Cloud Computing?
- The need for cloud computing in modern businesses.
- Why organizations moved to Cloud
- Cloud Service Providers – GCP, Azure, AWS, ...

02 GCP Cloud Basics

7 TOPICS

- GCP Cloud basics
- Introduction to Google Cloud Platform (GCP) and its core services
- GCP interfaces - Console, Cloud Shell and Cloud SDK
- GCP Locations - Regions, Zones & Global Infrastructure
- GCP Walkthrough - GCP Projects, Organizations, Folders & Billing Accounts
- GCP IAM & Admin - Principals, Roles, Service Accounts & Resource Hierarchy
- Setting up a Free Account in GCP

03 Linux Basics on Cloud Shell

8 TOPICS

- Getting started with Linux
- Linux Installation
- Basic Linux Commands
- Cloud shell tips
- File and Directory Operations (ls, cd, pwd, mkdir, rmdir, cp, mv, touch, rm, nano)
- File Content Manipulation (cat, less, head, tail, grep)
- Text Processing (awk, sed, cut, sort, uniq)
- User and Permission related (whoami, id, su, sudo, chmod, chown)

04 Python for Data Engineer

25 TOPICS

- **Getting Started with Python**
- Introduction to Python and basic functions
- Data Types – Strings, Operators, Numbers (Int, Float), Booleans
- Data Structures – Lists, Tuples, Dictionaries, Sets
- Python Programming Constructs - if, elif, else statements, for loops, while loops
- File I/O operations - Read, write, append, rename, remove, copy, move
- **Exception Handling and Logging**
- what is exception handling, why it is needed
- try and except blocks, else, finally, raise blocks
- Why Logging? And Logging Levels
- Writing Logs to File
- Data and Time operations - Working date and time, Formatting, Parsing, Time delta
- **Modular Programming in Python**
- Functions & Lambda Functions
- Classes
- **REST API basics**
- What is API
- HTTP methods (GET, POST, PUT, DELETE)
- Status Codes, Request Headers, JSON Responses
- Using the requests Library, Reading API Responses
- **Modules and Packages**
- What is module and packages
- Importing
- Built-in modules
- Install packages

02 GCP Data Engineering Tools

Storage, warehousing, batch, streaming & orchestration on Google Cloud

05 Google Cloud Storage

11 TOPICS

- Overview of Cloud Storage as a scalable and durable object storage service.
- Understanding buckets and objects in Cloud Storage.
- Use cases for Cloud Storage, such as data backup, multimedia storage, and website content
- **Labs: using console & CLI to do below**
- Creating and managing Cloud Storage buckets.
- Uploading and downloading objects to and from Cloud Storage.
- Setting access controls and permissions for buckets and objects.
- Data Transfer and Lifecycle Management
- Versioning and Object Versioning
- Integration with Other GCP Services
- Monitoring and logging for Cloud Storage operations.

06 Cloud SQL

12 TOPICS

- Introduction to Cloud SQL
- Creating and Managing Cloud SQL Instances
- Configuring database settings, users, and access controls.
- Connecting to Cloud SQL instances using Cloud SQL studio, Shell, Workbenches
- Importing and exporting data in Cloud SQL.
- Backups and High Availability
- Integration with Other GCP Services
- Managing database user roles and permissions.
- Introduction to DMS
- **End to End Database migration Project**
- Manual: Export and Import method
- Automation: Cloud SQL DMS method

07 BigQuery (SQL development)

18 TOPICS

- Introduction to BigQuery
- BigQuery Architecture
- Use cases for BigQuery in business intelligence and analytics.
- Various method of creating table in BigQuery
- BigQuery Data Sources and File Formats
- Native table and External Tables
- Working with complex data types
- Working json data, nested, repeated and array data
- **Data Integration and Export**
- Loading data into BigQuery from Cloud Storage, Cloud SQL, and other sources.
- Exporting data from BigQuery to various formats.
- Real-time data streaming into BigQuery.
- Configuring access controls and permissions in BigQuery.
- **BigQuery Views: Views, Materialized Views, Authorized Views**
- Optimization techniques in BigQuery
- BigQuery Slots – on demand, flat-rate, flex-slots
- Case Study-1: implement a real-world analytics data platform for Spotify
- Case Study-2: Enterprise Social Media analytics platform

08 DataProc (Pyspark Development)

19 TOPICS

- Introduction to Hadoop and Apache Spark
- Understanding the difference between Spark and MapReduce
- What is Spark and Pyspark.
- Understanding Spark framework and its functionalities
- Overview of DataProc as a fully managed Apache Spark and Hadoop service.
- **Cluster Creation and Configuration**
- Creating and managing DataProc clusters.
- Configuring cluster properties for performance and scalability.
- Preemptible instances and cost optimization.
- **Learning Pyspark**
- How to read from multiple data sources – csv, text, json, parquet, database table, BigQuery tables
- How to perform multiple transformations
- How to write to multiple targets - csv, text, json, parquet, database table, BigQuery tables
- **Running Jobs on DataProc**
- Submitting and monitoring Spark and Hadoop jobs on DataProc.
- Use of initialization actions and custom scripts.
- Job debugging and troubleshooting.
- Case study-1: Data Cleaning of Employee Travel Records
- Case study-2: Processing real-time patient health data
- Case study-3: Creating a pyspark job to support ML model creations

09 DataFlow (Apache Beam development)

11 TOPICS

- Introduction to DataFlow
- Use cases for DataFlow in real-time analytics and ETL.
- Understanding the difference between Apache Spark and Apache Beam
- How Dataflow is different from Dataproc
- **Learning Apache Beam**
- How to read from multiple data sources – csv, text, json, parquet, database table, BigQuery tables
- How to perform multiple transformations
- How to write to multiple targets - csv, text, json, parquet, database table, BigQuery tables
- Case study-1: Template method of creating pipelines
- Case study-2: E-commerce Transaction Processing
- Case study-3: End to End Streaming Pipeline using Apache beam with Dataflow, Python app, PubSub, BigQuery, GCS

10 Cloud Pub/Sub (Streaming)

10 TOPICS

- Introduction to Pub/Sub
- Understanding the role of Pub/Sub in event-driven architectures.
- Key Pub/Sub concepts: topics, subscriptions, messages, and acknowledgments.
- **Creating and Managing Topics and Subscriptions**
- Using the GCP Console to create Pub/Sub topics and subscriptions.
- Configuring message retention policies and acknowledgment settings.
- **Publishing and Consuming Messages**
- Writing and deploying code to publish messages to a topic.
- Implementing subscribers to consume and process messages from subscriptions.
- Case study-1: Streaming use-case using Dataflow

11 Cloud Composer (DAG Development)

19 TOPICS

- Introduction to Composer/Airflow
- Overview of Airflow Architecture
- Use cases for Composer in managing and scheduling workflows.
- **Creating and Managing Workflows**
- Creating and configuring Composer environments.
- Defining and scheduling workflows using Apache Airflow.
- Monitoring and managing workflow executions.
- **Integration with Data Engineering Services**
- Orchestrating workflows involving BigQuery, DataFlow, and other services.
- Coordinating ETL processes with Composer.
- Integrating with external systems and APIs.
- **Error Handling and Troubleshooting**
- Handling errors and retries in Composer workflows.
- Debugging and troubleshooting failed workflow executions.
- Logging and monitoring for Composer workflows.
- Level-1-DAG: Orchestrating the BigQuery pipelines
- Level-2-DAG: Orchestrating the DataProc pipelines
- Level-3-DAG: Orchestrating the Dataflow pipelines
- Deploy DAGs: Implementing CI/CD in Composer Using Cloud Build and GitHub

03 Azure Data Engineering Tools

Azure platform, data lake, factory & warehouse services

12 Azure Introduction

6 TOPICS

- Introduction to Azure Cloud Platform and its core services
- Azure interfaces - Console, Cloud Shell and Cloud SDK
- Azure Locations - Regions, Zones & Global Infrastructure
- Azure Walkthrough – Subscriptions, Resources, Resource group & Management group
- Azure IAM & Admin - Identities, Roles, Service Accounts & Resource Hierarchy
- Setting up a Free Account in Azure Cloud

13 Azure DataLake Storage (ADLS)

21 TOPICS

- Introduction to Azure Data Lake Storage (ADLS)
- Core Components of ADLS Gen2
- Blobs, Containers, and Storage Accounts
- Hierarchical Namespace
- **Why Use Azure Data Lake Storage?**
- Scalability, Cost-Effectiveness, Security
- Integration with Analytics Services
- **Access and Management of ADLS**
- Data Management via Console, SDK, CLI, REST API
- Role-Based Access Control (RBAC) and Access Control Lists (ACLs)
- **Pricing Overview**
- Pricing Tiers (Hot, Cool, Archive)
- Cost Factors
- **Best Practices for ADLS**
- Data Partitioning
- Tiering Strategy for Cost Optimization
- Monitoring and Usage Tracking
- **Interactive Access Methods**
- Managing ADLS via Console
- CLI/Shell Interaction
- Programmatic Access via SDKs

14 Azure SQL DB

3 TOPICS

- Introduction to Azure SQL Database
- Key Features
- Creating a SQL db for practice

15 Azure Data Factory (ADF)

20 TOPICS

- What is ADF and Use cases
- Core components of ADF and understanding & configuring
- Integration runtime
- Linked Services
- Datasets
- Pipelines
- Activities
- Triggers
- Key Features of Azure Data Factory
- **Pipelines in ADF**
- Creating and Configuring Pipelines
- Activities and Triggers
- Monitoring and Managing Pipeline Runs
- Orchestration
- Multiple examples with ADF activities
- Case study-1: End to End Incremental pipeline with ADF, ADLS, SQL DB.
- **Best Practices for ADF**
- Designing Efficient Pipelines
- Data Partitioning and Parallelism
- Monitoring and Optimization for Performance

16 Azure Synapse

5 TOPICS

- Synapse Workspace, SQL Pools & Architecture
- Dedicated & Serverless SQL concepts
- Data ingestion using Synapse Pipelines
- External Tables, Views & Data Lake integration
- Query performance optimization, partitioning & workload management

17 Databricks Lakehouse Platform

23 TOPICS

- Introduction to Databricks Lakehouse Platform
- **Databricks Architecture and Core Components**
 - Explore the architecture, including clusters, jobs, and the workspace environment.
 - Key components like Databricks Runtime, Spark, and Delta Lake.
- **Setting Up and Administering a Databricks Workspace**
 - Step-by-step guide on workspace creation and management.
 - Admin tasks including user management, permissions, and billing.
- **Working with Notebooks and Clusters**
 - Collaborative data science using Databricks notebooks.
 - Cluster setup, management, and autoscaling.
- **Databricks Unity Catalog**
 - Unified governance solution for managing data and metadata across the lakehouse.
 - Organizing data assets and enforcing security policies.
- **Managing Data with Delta Lake**
 - Understanding Delta Lake's ACID transactions and versioning capabilities.
 - Best practices for efficient data management.
- Creating Delta Lake Tables
- Querying and managing data across multiple file formats.
- Building ELT Pipelines with Spark SQL and Python
- Incremental Data Processing with Delta Lake
- Case Study 1: Real-time data processing with Spark Structured streaming
- Case Study 2: Databricks Autoloader for file-based ingestion utility
- Performance Optimization in Databricks

PROGRAM OUTCOMES

By the End of the Course

What students can expect to achieve on completion of the Multi-Cloud Data Engineer Program.

01

Build end-to-end Data Engineering pipelines on GCP & Azure

02

Develop strong skills in SQL, Python, PySpark, Linux & Apache Beam

03

Work with industry standard data engineering services

04

Implement Batch & Real-Time Data Processing pipelines

05

Design Bronze → Silver → Gold / Medallion Architecture

06

Build production-ready projects and troubleshoot real-world data engineering scenarios

07

Understand Cloud, Data Lake, Data Warehouse & Lakehouse architectures

08

Prepare for Data Engineer interviews with practical, scenario-based knowledge

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Thank You.

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Start your multi-cloud data engineering journey with **SkillVane IT Academy**
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